

Table 1: Performance comparisons of Official X-Fi (Image-based) vs. Ours (Privacy-Preserving Keypoint-based) on the MM-Fi dataset. Error reductions (improvements) are marked in **blue**, while error increases (degradations) are marked in **red**. ‘VK’ denotes 2D Visual Keypoints replacing the raw RGB modality.

Modality	MPJPE ↓ (mm)			PA-MPJPE ↓ (mm)		
	Official X-Fi (I)	Ours (VK)	Imp↑	Official X-Fi (I)	Ours (VK)	Imp↑
I/VK	93.9	120.2	-28.0%	60.3	50.5	+16.3%
D	101.8	80.8	+20.6%	48.4	48.8	-0.8%
L	167.1	152.0	+9.0%	103.2	89.6	+13.2%
R	127.4	117.5	+7.8%	69.8	57.8	+17.2%
W	225.6	205.5	+8.9%	105.3	106.2	-0.9%
I/VK+D	86.1	78.1	+9.3%	48.1	46.8	+2.7%
I/VK+L	93.0	103.5	-11.3%	59.7	49.4	+17.3%
I/VK+R	88.8	94.0	-5.9%	57.3	44.4	+22.5%
I/VK+W	93.0	112.7	-21.2%	59.5	48.7	+18.2%
D+L	102.5	81.2	+20.8%	48.4	48.6	-0.4%
D+R	98.0	78.7	+19.7%	47.3	47.2	+0.2%
D+W	101.8	80.0	+21.4%	48.1	48.3	-0.4%
L+R	109.8	99.5	+9.4%	63.4	53.9	+15.0%
L+W	159.5	120.5	+24.5%	102.7	78.6	+23.5%
R+W	117.2	108.4	+7.5%	62.7	57.4	+8.5%
I/VK+D+L	84.8	76.6	+9.7%	48.2	46.7	+3.1%
I/VK+D+R	83.4	74.8	+10.3%	47.3	45.1	+4.7%
I/VK+D+W	85.3	76.3	+10.6%	48.1	46.1	+4.2%
I/VK+L+R	88.4	90.6	-2.5%	57.2	44.6	+22.0%
I/VK+L+W	93.0	94.8	-1.9%	59.7	47.7	+20.1%
I/VK+R+W	88.5	90.5	-2.3%	57.1	43.5	+23.8%
D+L+R	96.0	79.8	+16.9%	47.3	47.9	-1.3%
D+L+W	102.0	79.9	+21.7%	48.1	48.4	-0.6%
D+R+W	97.0	78.0	+19.6%	47.1	47.3	-0.4%
L+R+W	107.4	94.4	+12.1%	63.1	53.3	+15.5%
I/VK+D+L+R	83.5	74.8	+10.4%	47.6	46.2	+2.9%
I/VK+D+L+W	86.0	75.0	+12.8%	48.2	46.7	+3.1%
I/VK+D+R+W	84.0	74.8	+11.0%	47.6	46.0	+3.4%
I/VK+L+R+W	88.6	86.5	+2.4%	57.1	44.0	+22.9%
D+L+R+W	97.6	78.0	+20.1%	47.4	47.4	0.0%
I/VK+D+L+R+W	83.7	73.2	+12.5%	47.6	45.8	+3.8%